Activity 2

Stand erect and wave your hands in side wards. Take a deep breath and feel your rib movements. Then run some 100 metres and observe the rib movements. Discuss in the class room about what you observed.

Activity 3

Constructing a model of lungs.

Materials required

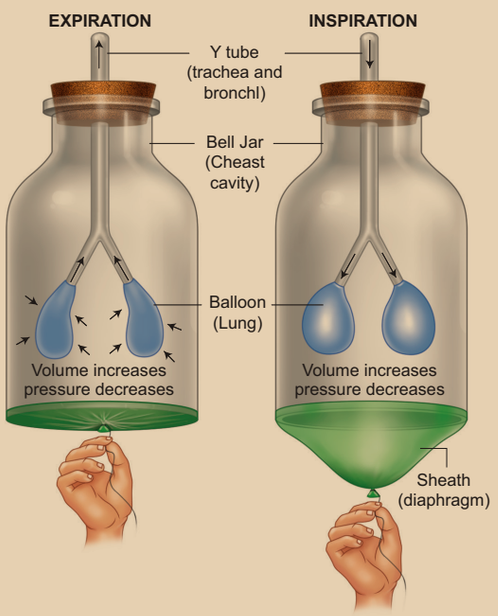
Y shaped tube, a large balloon, two small balloons, a one litre plastic bottle, cork.

Method of Construction

Cut off the plastic bottle in the middle. Fix two small balloons in both the ends of the Y-tube. Make a hole in the cork and fix the y-tube. Make a small hole in the cork and fix the y-tube through the hole as shown in the picture. Cut a large balloon into two halves and fix one half tightly around the open part of the bottle.

Method of Working

Hold the large balloon in the middle and pull it slowly downwards as shown in the picture. Observe the change in the balloons inside the bottle. Now leave the balloon free.



18.6 Physiological Processes

The ways in which biomolecules, cells, tissues, organs and organs systems work together to accomplish the complex goal of sustaining life are called physiological processes. Let us study about some of them here.

18.6.1 Homeostasis

Homeostasis is a property of human biological system where the **self-regulating** process tends to maintain the balance for the survival. The regulation takes place in a defined internal environment. Mammals are capable of maintaining constant body temperature despite the changes in the external temperature. Behavioural and physiological responses are the two important regulating mechanisms that maintain the stability of homeostasis.

In simple terms, it could be referred as a balance in a system to maintain a stable internal environment for the survival of the animal. If the homeostasis regulates successfully, life continues or if unsuccessful, death or disaster occurs.

All the processes of integration and co-ordination of function are mediated by nervous and hormonal system. The liver, kidney, and brain (hypothalamus), autonomic nervous system and the endocrine system help to maintain homeostasis.

Maintenance of body fluid concentrations, body temperature are done by various bio-physical and bio-chemical methods. Human beings are warm blooded in nature i.e, they maintain their body temperature as constant. When the body temperature raises sweat is produced to bring the temperature down. When the body temperature lowers heat is produced by the muscular work by shivering. This is an example for homeostasis.

The control of blood glucose level is another example in which insulin hormone is secreted whenever the blood glucose level raises and glucagon hormone is secreted whenever the blood glucose level reduces.

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